

Group-index extraction with unbalanced silicon Mach–Zehnder interferometers

Jakob Hinum-Wagner

Graz University of Technology

UBCx Phot1x: Silicon Photonics Design, Fabrication and Data Analysis

edX Public Username: indium4arsenid

Abstract—This proposal describes eight unbalanced Mach–Zehnder interferometers for extracting the group index of a 500 nm-wide, 220 nm-high silicon strip waveguide in quasi-TE polarisation. The primary comparison is between the group index obtained from the measured free spectral range and that calculated in a waveguide mode solver. Two path-length differences, 40 and 60 μm , are combined with two mean arm lengths, 130 and 160 μm , with two devices per combination. Two loopbacks provide reference measurements. The additional uncertainty study separates spectral measurement error from device-to-device variation and retains correlation between the arm responses. Geometry and analytical design checks are documented here; the required mode-solver exports and their simulation-derived plots are not yet available in this revision.

Index Terms—Silicon photonics, Mach–Zehnder interferometer, group index, free spectral range, measurement uncertainty.

I. INTRODUCTION

The design objective is to measure the waveguide group index from an interferometer spectrum and compare it with a simulation of the same waveguide, as required by the Phot1x proposal brief [1]. I use the default strip waveguide and Y-branch splitter so that this comparison does not depend on an additional component-development task. A quasi-TE input and a nominal wavelength of 1550 nm are used throughout; results from a quasi-TM mode would not describe this layout.

The first question is whether both path-length differences give the same group-index estimate within the measurement uncertainty. The second is whether nominally repeated circuits agree. Mean arm length is varied at fixed path-length difference to examine effects which depend on the complete arm paths, rather than on the imbalance alone. This uncertainty study is an extension of the course objective, not a substitute for the mode simulation or the group-index comparison.

The source defines ten measurable circuits in a $605 \times 410 \mu\text{m}$ tile: eight MZIs and two loopbacks [2]. There are four distinct MZI parameter combinations, each repeated twice; the loopbacks are references, not two additional interferometer designs. The brief permits repeated designs for a variability comparison [1].

Status of this revision. No matching quasi-TE mode-profile or wavelength-sweep export was included with the report sources. Sections III and III-E therefore distinguish the calculation method from missing simulation evidence. In particular, the numerical index used for the illustrative FSR check is an assumption, not a MODE result.

TABLE I
SUBMISSION IDENTITY AND NOMINAL WAVEGUIDE

Public username	indium4arsenid
Report	EBeam_indium4arsenid.pdf
Layout	EBeam_indium4arsenid.gds
Fork	indium4arsenid/openEBL-2026-10
Cross-section	Fully etched silicon strip; width 500 nm, height 220 nm
Polarisation	Fundamental quasi-TE mode
Nominal wavelength	1550 nm

II. THEORY

A. Propagation and waveguide compact model

For a guided mode with vacuum wavelength λ , the real propagation constant is

$$\beta(\lambda) = \frac{2\pi}{\lambda} n_{\text{eff}}(\lambda). \quad (1)$$

Here n_{eff} denotes the real part of the modal effective index. The group velocity and group index are [3]

$$v_g = \left(\frac{d\beta}{d\omega} \right)^{-1}, \quad n_g = c \frac{d\beta}{d\omega} = n_{\text{eff}} - \lambda \frac{dn_{\text{eff}}}{d\lambda}. \quad (2)$$

The proposed compact model is a local quadratic fit,

$$x = \frac{\lambda - \lambda_0}{s}, \quad \lambda_0 = 1.55 \mu\text{m}, \quad s = 0.025 \mu\text{m},$$

$$n_{\text{eff}}^{\text{fit}}(\lambda) = b_0 + b_1 x + b_2 x^2, \quad (3)$$

$$n_g^{\text{fit}}(\lambda) = b_0 + b_1 x + b_2 x^2 - \frac{\lambda}{s} (b_1 + 2b_2 x). \quad (4)$$

The coefficients b_j are dimensionless because x is dimensionless. This convention avoids an implicit nm-to- μm conversion in the derivative. The same wavelength units must be used in λ , λ_0 and s . Numerical coefficients are to be fitted to the actual MODE sweep, not selected to obtain a desired group index. The fit is valid only over the simulated wavelength interval.

With a power-attenuation coefficient α_P , the field transfer of a uniform length L is

$$h_L(\lambda) = \exp[-\alpha_P(\lambda)L/2] \exp[i\beta(\lambda)L]. \quad (5)$$

If attenuation is specified in dB/cm, its value in μm^{-1} is $\alpha_P = (\ln 10/10) \alpha_{\text{dB/cm}} 10^{-4}$. The factor of two in the field exponent must not be confused with the power-attenuation convention. No measured propagation-loss value is available for this tile.

B. Interferometer transfer function

Let $a_s(\lambda)$ and $a_l(\lambda)$ include the complex splitter/combiner coefficients, access paths and grating response for the short and long arms. The transmission is

$$T(\lambda) = |a_s h_{L_s} + a_l h_{L_l}|^2. \quad (6)$$

For ideal equal-amplitude splitting and combining, with common propagation loss, the on-chip normalised expression becomes

$$T_{id}(\lambda) = \frac{1}{4} [e^{-\alpha_P L_s} + e^{-\alpha_P L_l} + 2e^{-\alpha_P (L_s + L_l)/2} \cos \Phi(\lambda)], \quad (7)$$

$$\Phi(\lambda) = \frac{2\pi n_{eff}(\lambda) \Delta L}{\lambda} + \phi_Y(\lambda), \quad (8)$$

$$\Delta L = L_l - L_s > 0.$$

The differential Y-junction phase is denoted by ϕ_Y . In the lossless, zero-differential-phase limit, $T_{id} = [1 + \cos(2\pi n_{eff} \Delta L / \lambda)]/2$. A measured fibre-to-fibre spectrum also contains coupling loss and a spectral envelope; a practical fit is $A(\lambda) + B(\lambda) \cos \Phi$, with $A \geq |B|$. This fit is not a substitute for a characterised splitter response.

C. Deriving the group-index estimator

Two adjacent maxima, or two adjacent minima, are separated by a phase change of 2π . A maximum and its neighbouring minimum are only half a fringe and must not be used as one FSR. Neglecting differential component dispersion for the first derivation,

$$\begin{aligned} \frac{d\Phi}{d\lambda} &= 2\pi \Delta L \left(\frac{1}{\lambda} \frac{dn_{eff}}{d\lambda} - \frac{n_{eff}}{\lambda^2} \right) \\ &= -\frac{2\pi n_g \Delta L}{\lambda^2}. \end{aligned} \quad (9)$$

Linearising over one fringe gives

$$FSR_\lambda \simeq \frac{\lambda^2}{n_g \Delta L}, \quad \boxed{\widehat{n_g} \simeq \frac{\lambda_c^2}{\Delta L FSR_\lambda}}. \quad (10)$$

All lengths in this equation must have consistent units.

A useful finite-interval form follows in optical frequency $\nu = c/\lambda$:

$$\frac{d\Phi}{d\nu} = \frac{2\pi \Delta L}{c} n_g(\nu). \quad (11)$$

Integrating between adjacent same-type extrema at $\lambda_a < \lambda_b$ gives

$$\boxed{\overline{n_g}[\nu_b, \nu_a] = \frac{c}{\Delta L |\nu_a - \nu_b|} = \frac{\lambda_a \lambda_b}{\Delta L (\lambda_b - \lambda_a)}}. \quad (12)$$

This is a frequency-interval average, not exactly the point value at the arithmetic wavelength midpoint. It is exact under the stated phase model and correct fringe order. Equation (10) follows by replacing $\lambda_a \lambda_b$ with λ_c^2 .

D. What the real circuit measures

More generally, for arm paths γ_l and γ_s ,

$$D_g = \int_{\gamma_l} n_g(\mathbf{r}, \lambda) ds - \int_{\gamma_s} n_g(\mathbf{r}, \lambda) ds - \frac{\lambda^2}{2\pi} \frac{d\phi_Y}{d\lambda}, \quad (13)$$

$$FSR_\lambda \simeq \frac{\lambda^2}{|D_g|}. \quad (14)$$

Dividing by ΔL gives the uniform-waveguide group index only when both arms share the relevant dispersion and the differential component term is negligible or corrected. Equal bend counts and radii help cancel nominal bend phase; they do not prove cancellation after fabrication.

III. MODELLING AND SIMULATION

A. Waveguide calculation and required outputs

The intended MODE calculation is a 2D cross-section of the fully etched 500×220 nm silicon strip, using the cladding and material model appropriate to the course EBeam stack. Oxide surroundings are the nominal modelling assumption for this oxide-clad design. Actual oxide thicknesses, material fits, simulation temperature and boundary placement must be recorded from the simulation project; they are not inferred from the GDS.

MODE's FDE solver provides the spatial mode fields and effective index, and its frequency analysis provides group velocity and dispersion [3], [4]. A 1525–1575 nm sweep is proposed to match an illustrative 50 nm measurement window centred at 1550 nm. The brief specifies approximately 50 nm bandwidth, not those exact endpoints. The usable experimental interval must be checked against the actual grating response.

At 1550 nm, the fundamental quasi-TE mode is to be identified by its field distribution and polarisation fraction, not by assuming that a mode number always denotes TE. The saved profile must show the cross-section, coordinate axes, wavelength and the displayed field quantity, for example normalised $|E|^2$. The same mode is then tracked through at least 51 wavelengths. The source's 1 nm GDS grid is not an optical mesh size.

The simulation record should include a mesh/domain comparison. Suggested starting comparisons are a 10 nm and 5 nm local mesh and an enlarged transverse domain. These are proposed checks, not converged settings. Recalculate n_{eff} and n_g and compare both; a small error in n_{eff} alone does not establish a reliable wavelength derivative.

B. Mode-profile and dispersion evidence still required

The required MODE profile and plots of $n_{eff}(\lambda)$ and $n_g(\lambda)$ are not yet included. The supplied attachments contain layout code and analytical-geometry tests, not the corresponding electromagnetic exports. The revised Overleaf project accepts these outputs through the supplied export and MATLAB processing scripts. Until those data are imported and checked, this remains a draft proposal rather than a complete simulation report.

C. Fitting and checking the compact model

The quadratic in (3) will be fitted to the real effective index. Alternating wavelengths are retained as a hold-out check; after that check the final coefficients use the complete sweep. The residual in effective index and the difference between (4) and the directly exported c/v_g are both reported. A higher-order polynomial is justified only if it improves these checks without introducing unstable curvature. Fit residuals from deterministic solver outputs are approximation errors, not automatically statistical confidence intervals.

Numerical model status: b_0 , b_1 and b_2 remain unfitted. They cannot be obtained from the waveguide geometry alone without an optical calculation. No assumed coefficients are presented as the compact model of the fabricated waveguide.

D. Circuit calculation and the simulation test

Once the waveguide data are available, the MATLAB calculation will evaluate (7) for $\Delta L = 40$ and $60 \mu\text{m}$. The initial spectra are normalised, lossless analytical-circuit simulations using the exported dispersion; they do not predict fibre-to-fibre insertion loss. A later INTERCONNECT model can replace the ideal Y coefficients with the corresponding component response.

The group-index test uses adjacent same-type extrema of those spectra in (12). The extracted interval-average n_g is compared with c/v_g from MODE averaged over the same frequency interval. This avoids confusing finite-FSR averaging with a simulation discrepancy. The MATLAB helper constructs the spectrum from an interpolation of the raw effective-index samples, rather than only from the same fitted polynomial used for the derivative comparison.

Circuit-result status: no spectrum driven by the actual MODE dispersion has been supplied. Its plot, fringe table and comparison with the direct MODE group index remain open requirements. The independent analytical check below does not close that gap.

E. Analytical consistency check, not a MODE result

To test the equations and units before importing optical data, I use an explicitly illustrative model

$$n_{\text{eff}}^{\text{demo}}(\lambda) = 2.5 - \frac{1.5}{1.55 \mu\text{m}}(\lambda - 1.55 \mu\text{m}), \quad (15)$$

which has $n_g = 4.0$ by construction. Neither 2.5 nor 4.0 is a simulated index for this submission. Loss and differential splitter phase are set to zero for this check.

The local FSR at 1550 nm is 15.0156 nm for $40 \mu\text{m}$ and 10.0104 nm for $60 \mu\text{m}$. Exact extrema of the analytical phase give representative adjacent-maximum pairs of approximately $1542.7683/1557.7889 \text{ nm}$ and $1547.7429/1557.7889 \text{ nm}$, respectively. Inserting the unrounded extrema into (12) returns 4.000000 for both cases to numerical precision. The numerical check and unrounded values are included in the project data. This tests the extraction formula, not the waveguide simulation, fabrication or measurement accuracy.

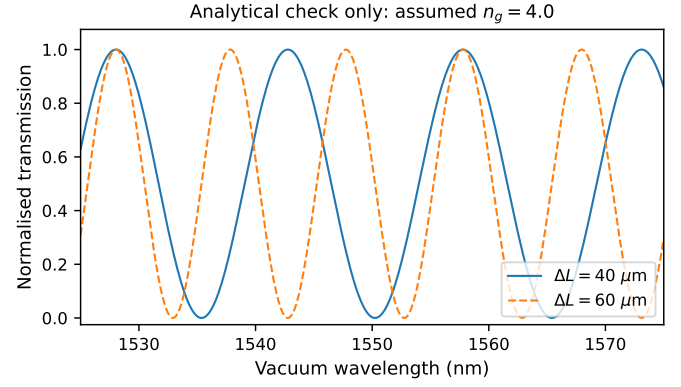


Fig. 1. Illustrative normalised spectra calculated from (15). The two path-length differences produce different fringe spacings. These are analytical test curves with an assumed group index, not MODE or INTERCONNECT results for the submitted device.

IV. DESIGN VARIANTS AND LAYOUT

A. Bandwidth check

For a measurement bandwidth B_λ , the local requirement $\text{FSR}_\lambda < B_\lambda$ gives

$$\Delta L > \frac{\lambda_0^2}{n_g B_\lambda}. \quad (16)$$

At $\lambda_0 = 1.55 \mu\text{m}$ and $B_\lambda = 0.05 \mu\text{m}$, the $40 \mu\text{m}$ choice satisfies this condition if $n_g > 1.20125$, and the $60 \mu\text{m}$ choice if $n_g > 0.80083$. With the explicit planning value 4.0 , both fit several fringe intervals within the nominal bandwidth. The ratio 3.33 or 5.00 is a local planning estimate, not a guaranteed integer count of visible fringes. At the edges of the band the local FSR and available signal must be checked again using the actual dispersion and grating transmission.

At fixed n_g , the predicted ratio $\text{FSR}_{40}/\text{FSR}_{60} = 60/40 = 1.5$ is a direct design-level check. Changing $\bar{L} = (L_s + L_1)/2$ without changing ΔL does not change the ideal FSR. It can change attenuation and sensitivity to spatially nonuniform perturbations.

B. Routing and component choices

Each MZI uses two ebeam_y_1550 cells and two GC_T E_1550_8degOxide_BB cells. These are the requested library names, not newly simulated splitter or grating designs [2], [5]. Each arm has two circular 90° bends of radius $10 \mu\text{m}$; each access route has two bends of radius $5 \mu\text{m}$. The lengths in Table II are between the Y-junction optical pins. They exclude the grating couplers, access paths and the internal Y-junction lengths.

The input–output grating pitch is $127 \mu\text{m}$. Column origins are $x = 60, 170, 280, 390, 500 \mu\text{m}$. The lower/upper grating coordinates are $y = 20/147 \mu\text{m}$ in the lower row and $240/367 \mu\text{m}$ in the upper row. The smallest origin separation is $93 \mu\text{m}$; this is not a statement about full-cell clearance. All four treatment pairs have the same mean grating-pair coordinate, $(280, 193.5) \mu\text{m}$. This balances an additive linear position effect at the chosen coordinates, not the complete routed footprints.

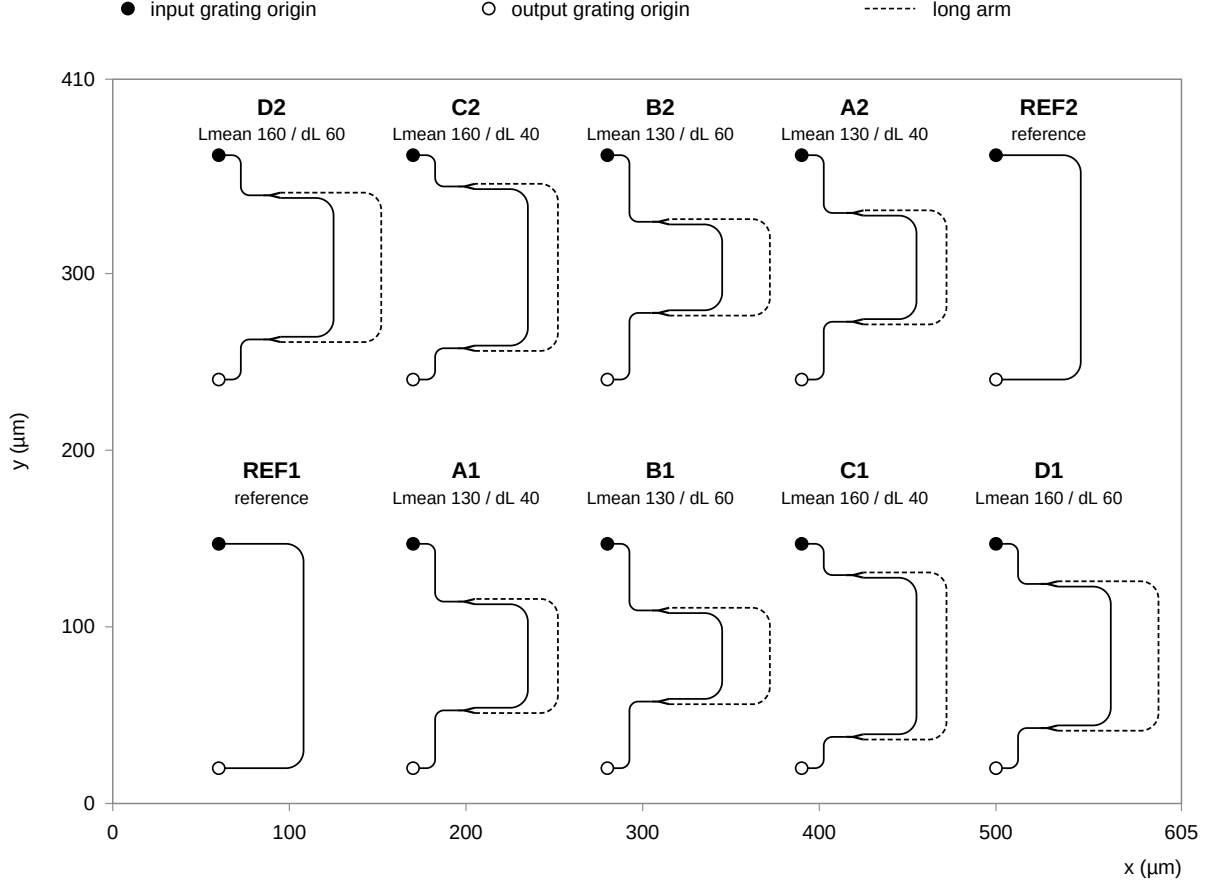


Fig. 2. Source-defined floorplan of the eight MZIs and two loopbacks. Filled and open circles mark the upper input and lower output grating origins. The centrelines use the generator’s assumed port map and zero clearance shift; splitter symbols and linewidths are not mask dimensions. The drawing must be checked against the final exported GDS and runtime geometry record.

TABLE II

NOMINAL VARIANTS AND CONDITIONAL EXPECTED PERFORMANCE AT 1550 NM. THE NUMERICAL FSR USES THE EXPLICIT PLANNING ASSUMPTION $n_g = 4.0$; THE EXACT FORMULA MUST LATER USE THE SIMULATED GROUP INDEX. ALL ARMS USE THE SAME 500 NM WIDTH, 220 NM HEIGHT AND QUASI-TE POLARISATION.

Variant	Device IDs	$\bar{L}$ (μm)	L_s (μm)	L_1 (μm)	ΔL (μm)	FSR (nm)	50 nm/FSR
A	A1, A2	130	110	150	40	15.02	3.33
B	B1, B2	130	100	160	60	10.01	5.00
C	C1, C2	160	140	180	40	15.02	3.33
D	D1, D2	160	130	190	60	10.01	5.00

Both loopbacks have the same geometry. They can monitor the spectral coupling envelope and drift but do not constitute a cutback series. Their measurements cannot by themselves separate propagation loss from coupling loss.

V. FABRICATION AND VERIFICATION RECORD

The intended submission file is EBeam_indium4arsenid.gds; the matching proposal filename is EBeam_indium4arsenid.pdf. This corrects the earlier report suffix _Design.pdf to the filename required by the supplied course brief [1]. No mask change is made by this report revision.

A. Clearance adjustment and the final geometry

The generator checks the actual installed component-recognition polygons and tries common routing shifts from 0 to 10 μm. A shift moves the splitters and exchanges horizontal and vertical straight lengths, preserving each arm’s analytical length, bend count, width and radius. It selects the first overlap-free candidate and then invokes SiEPIC functional verification [2]. This is a bounded clearance search, not optical optimisation.

The selected shift is not recorded in the supplied local test file. A nonzero shift changes spatial covariance integrals even when ΔL remains unchanged. The final geometry.json is therefore required before carrying out the layout-aware uncertainty calculation. The figures remain source reconstructions.

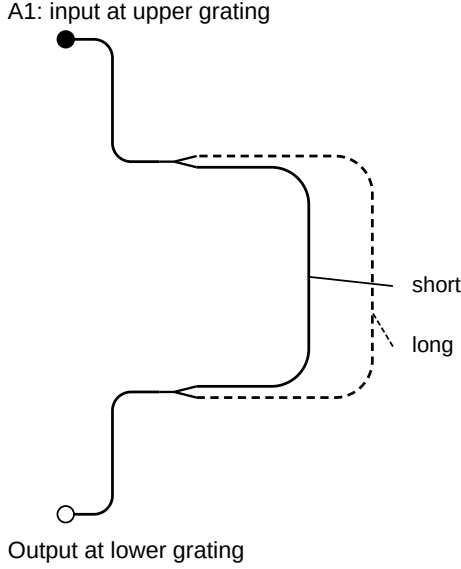


Fig. 3. A1 centreline example with a $110\mu\text{m}$ short arm and a $150\mu\text{m}$ long arm. The upper grating is the input. This is an analytical reconstruction with assumed ports, not a drawing of the library-cell polygons.

B. Evidence available for this proposal

The supplied `LOCAL_TEST_STATUS.json` records seven passing source/analytical-geometry tests, covering 12 assumed splitter port maps and 11 shifts. It explicitly records actual-PDK loading, KLayout GUI execution, functional verification, Photix design-for-test checks, manufacturing DRC and optical validation as not run in that test environment [6]. This is not evidence that a subsequent run on another computer failed.

The GDS export code requires zero local overlap and functional errors. It preserves hierarchy and the 1 nm database unit and checks the saved black-box name and bounds. Its 3 nm analytical-length and 20 nm rendered-delay tolerances are software checks, not fabrication tolerances. Run-specific `build_status.json`, final-mask checks and the merged-layout inspection remain to be attached. No current GitHub merge or manufacturing-pass status is claimed here.

Fabrication process details, realised dimensions and any departures from the intended design will be added after fabrication. They are not required experimental results at the proposal stage.

VI. EXPERIMENT DATA: PLANNED ACQUISITION

Experimental data will be added after the course measurements. The initial deliverable is a wavelength-calibrated TE transmission spectrum for each of the ten circuits, with device ID, die ID, session, alignment and sweep identifiers retained separately.

The approximately 50 nm scan must contain at least two adjacent same-type extrema with adequate signal. A sampling interval much smaller than the expected 10–15 nm FSR is needed; the final choice should follow a numerical sampling study and the instrument limits rather than an arbitrary number

of points. The actual usable window, optical power and detector background will be documented.

REF1 and REF2 will be measured before and after MZI blocks where the course measurement access permits this. MZI order should change between blocks to avoid tying a variant to a fixed measurement time. A proposed pilot is five sweeps at each of three independent alignments per device. These are requested measurement controls, not capabilities guaranteed by the automated course test service. If only one sweep is supplied, repeatability and alignment variance will remain unmeasured.

The raw spectra, wavelength calibration, temperature record and instrument settings will be retained. Reference division may remove part of a slowly varying envelope, but different grating pairs prevent exact de-embedding by an arbitrary loopback. Process claims beyond this tile require independent fabrication units rather than more wavelength samples from the same device.

VII. ANALYSIS AND UNCERTAINTY

A. Extracting and comparing the group index

The first analysis locates adjacent maxima and, independently, adjacent minima after checking the baseline and signal level. Extrema are refined locally rather than read from pixel positions in a figure. Envelope-induced peak shifts, reflections and missing fringes are checked against a full-spectrum phase fit. Every accepted pair is converted to a frequency-interval average by (12); the nominal path-length difference is replaced by the final design record where available.

The corresponding MODE index is evaluated over the same interval. A table will report $\widehat{n}_{\text{g,exp}}$, $\overline{n}_{\text{g,MODE}}$, their difference and the associated uncertainties for each device and wavelength interval. Agreement will not be judged solely from rounded values or a high fit R^2 . The 40 and 60 μm groups provide a cross-check; a systematic dependence on imbalance or mean length would prompt examination of calibration, bend/component dispersion or arm nonuniformity.

B. Uncertainty of a single extracted index

For $a = \lambda_a$, $b = \lambda_b$ and $D = \Delta L$, write

$$g(a, b, D) = \frac{ab}{D(b-a)}. \quad (17)$$

With all lengths expressed in the same units, its sensitivity vector is

$$\mathbf{J} = \left(\frac{b^2}{D(b-a)^2}, -\frac{a^2}{D(b-a)^2}, -\frac{g}{D} \right). \quad (18)$$

A first-order covariance propagation is [7]

$$u_c^2(g) = \mathbf{J}\mathbf{\Sigma}\mathbf{J}^T + u_{\text{model}}^2, \quad (19)$$

where $\mathbf{\Sigma}$ includes correlations between the wavelength estimates and path-length knowledge. The separate model term is appropriate only for contributions not already included and treated as independent in this expression. For nonlinear or appreciably non-Gaussian effects, propagation of input distributions is preferable [8].

A shared wavelength offset cancels from $b - a$, but not from the product ab . Adjacent FSR estimates also share peak locations, so they are not independent observations. Shared calibration errors do not shrink by averaging scans. Geometry grid tolerances are not a calibrated distribution for the fabricated ΔL . Similarly, solver mesh and material-model errors must be assessed separately from experimental fitting uncertainty.

C. Layout-aware process variation

The additional process study retains the arm paths rather than assigning independent random indices to the two arms. Layout-dependent process models already exist [9], [10]; their reported statistics are not transferred to this EBeam tile as though they were local calibration data.

For an effective-index perturbation $\delta n(\mathbf{r}, \lambda)$ at fixed wavelength,

$$\delta\Phi = \frac{2\pi}{\lambda} \left[\int_{\gamma_1} \delta n ds - \int_{\gamma_s} \delta n ds \right], \quad (20)$$

$$I_{ab} = \int_{\gamma_a} \int_{\gamma_b} C_n(\mathbf{r}, \mathbf{r}') ds ds', \quad (21)$$

$$\text{Var}(\delta\Phi) = \left(\frac{2\pi}{\lambda} \right)^2 (I_{11} + I_{ss} - 2I_{1s}). \quad (22)$$

An independent-arm comparison must keep the marginal variances and remove only the cross term. A uniform perturbation leaves $\delta\Phi_c = (2\pi/\lambda)\Delta L \delta n_c$; imbalance prevents complete common-mode cancellation.

Equation (22) is a phase-variation model at a single wavelength, not yet a group-index uncertainty model. For the latter, the spectral dependence must be retained:

$$\delta D_g = -\frac{\lambda^2}{2\pi} \frac{\partial \delta\Phi}{\partial \lambda}. \quad (23)$$

This requires correlated wavelength-dependent sensitivities or a covariance model for the group-index perturbation itself. Independently drawing a random phase at each wavelength would destroy the physical dispersion relationship and corrupt an FSR uncertainty estimate.

The changes in $\bar{L}$ and ΔL do not isolate every process mechanism: the routing also changes arm separation and splitter spacing. Two repeats per variant support a pilot comparison, not identification of a wafer-scale covariance function. Width, thickness and temperature will not be claimed as separately recovered merely because a multivariable fit converges.

VIII. CONCLUSION AND REMAINING PROPOSAL WORK

The source-defined design supports the course's group-index experiment: it uses a fixed 500×220 nm quasi-TE strip and two imbalances with an explicit FSR-to-bandwidth check. Repeated variants add a limited test of reproducibility without changing the waveguide cross-section. The group-index estimator, its assumptions and a numerical analytical check have been specified.

The simulation portion is not complete until the matching mode profile, effective/group-index sweeps, fitted polynomial coefficients and dispersion-driven circuit spectra are imported and reviewed. These are proposal requirements, whereas

experimental results and the final simulation-to-measurement discussion belong to the later report update. The uncertainty work should strengthen that comparison, not obscure the missing optical evidence.

PREPARATION AND SOURCE RECORD

The layout description is based on the supplied Build_Phot1x_UQ_OverlapAware.py, version 2.2.0-overlap-aware, and its local test record. The figures reconstruct the source-defined centrelines; they are not images of the final GDS. AI assistance was used in preparing the scripts and report. Numerical claims are separated below into design calculations, illustrative checks, and results still to be obtained.

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